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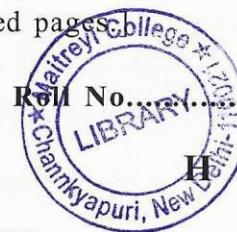
- (c) Multi drug resistant bacteria
- (d) Role of prebiotics in human health
- (e) Innate immunity

(1000)

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[This question paper contains 4 printed pages]

Your Roll No.....



Sr. No. of Question Paper : 5915

Unique Paper Code : 2234002005

Name of the Paper : GE: Microbiota: Importance
in Health & Disease

Name of the Course : B.Sc. (H) Zoology (NEP)

Semester : IV

Duration : 2 Hours

Maximum Marks : 60

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Attempt **any FOUR** questions in all.
3. **Question No. 1** is compulsory

1. (a) Define the following terms (1×6=6)

(i) Adaptive immunity

(ii) Gut-Microbiota

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(iii) Periplasmic space

(iv) Bioterrorism

(v) Virulence

(vi) Granuloma

(b) Distinguish between the following (1.5×4=6)

(i) Resident flora and Transient flora

(ii) Probiotics and Prebiotics

(iii) Humoral immunity and Cell mediated immunity

(iv) Chicken pox and Shingles

(c) Fill in the blanks (0.5×6=3)

(i) The fungi _____ is used for the production of first commercial antibiotic _____.

(ii) Methanogens are _____.

(iii) T_H and T_c cells are related to _____ immunity.

(iv) After encounter with antigen, B cell becomes _____ and _____ plasma cell.

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2. (a) Give a detailed account of the various bacterial diseases of the skin. (12)

(b) Write a short note on trichiasis (3)

3. (a) Diagrammatically explain the cell mediated response against bacteria. (8)

(b) Explain the bacterial endotoxins and exotoxins (7)

4. Give a brief description of the microbe that causes Tuberculosis. Give a detailed account of the symptoms, test, diagnosis, vaccine and preventive measures for the disease. (15)

5. Give an account of the following diseases and their causative agents. (7.5×2=15)

(a) Syphilis

(b) Tetanus

6. Write short note on any three (5×3=15)

(a) Factors affecting the children's gut microbiota

(b) Role of gut microbiome in GI cancer

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